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Day- 6
Introduction to Network Devices

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Q1. Create a table listing the main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home WiFi setup.

Answer:

Device	Main functions	Typical ports (home WiFi setup)
Switch	Connects wired devices (PC, printer, TV) in a LAN. It learns MAC addresses and sends data only to the correct port.	4 to 48 Ethernet ports, an uplink port to the router, power socket, and a console port on managed switches.
Router	Connects the home network to the Internet. Gives IP addresses (DHCP), uses NAT, chooses the path for data. Home routers usually include WiFi, a switch and a basic firewall.	1 WAN/Internet port, 4 LAN Ethernet ports, USB port, power socket, WiFi antennas, reset/WPS button.
Hub	Repeats the data it receives to all its ports. It is simple and slow, and is rarely used today.	4 to 8 Ethernet ports and a power socket.
Firewall	Checks network traffic and allows or blocks it using security rules, to protect the network from attacks.	WAN port, LAN port(s), DMZ port, console port, power socket. At home, it is usually built into the router.
Access Point	Creates the WiFi signal so wireless devices (phones, laptops) can join the wired network.	1 Ethernet (LAN/PoE) port, power socket (DC), antennas, reset button.

Q2. Given images or diagrams of network devices, label each device and mark the ports (Ethernet, console, power, antenna) for at least three devices.

Answer:



- **Router:** WAN/Internet port, LAN ports 1-4 (Ethernet), USB port, power socket, reset button and WiFi antennas.
- **Switch:** Ethernet ports 1-8, status LEDs, console port (for setup) and power socket.
- **Access Point:** Ethernet (LAN/PoE) port, power (DC) socket, reset button, status LED and antennas.
- **Firewall:** WAN, LAN and DMZ ports, console port, USB port and power socket.

Q3. Pick a popular app you use daily (like Instagram, Zomato, or Paytm) and describe in 3-4 lines how at least two different network devices (from switches, routers, firewalls, access points) play a role when you use that app.

Answer:

App: Zomato (placing a food order). When I tap "Place Order", my phone sends the data over WiFi to the access point, which connects it to the wired network. The access point passes it through a switch to the router. The router sends the order out to the Internet towards Zomato's server, and the reply comes back along the same path. The firewall (built into the router) checks the traffic and blocks anything unsafe, so my order and payment details stay protected.

Q4. Suppose your college WiFi is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason for the issue and how upgrading one device could help.

Answer:

Possible reason: Too many students are connected to one old access point (for example a single 802.11n access point serving a whole classroom). Its capacity is limited, so it becomes overloaded, speeds drop and some devices get disconnected.

Upgrade that can help: Replace it with a newer WiFi 6 access point (or add more access points), because WiFi 6 handles many devices at once and gives better coverage. The switch connected to it should have gigabit (and PoE) ports so that it does not become a bottleneck, and the router should be able to handle the traffic of all the users.